Mini Project (22UEE528P)
Report on

Design and Development of Boost converter for SPV system

Submitted in the partial fulfilment for the award of degree in
Bachelor of Engineering in the
Department of Electrical and Electronics Engineering

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Department of Electrical and Electronics Engineering

Certificate

This is to certify that Mini Project entitled "**Design and Development of Boost Converter for SPV system**" is a bonafide work of;

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The report satisfies the academic requirements in respect of project work prescribed for V semester **Mini Project (22UEE528P)** for award of Bachelor of Engineering degree in the department of Electrical and Electronics Engineering during the academic year 2025-2026.

Hereby declare that the submitted mini project work report is not submitted elsewhere.

Dr. R. L. Naik	**Prof. S. S. Kandagal**	**Dr. R. L. Naik**	**Dr. B. R. Hiremath**
Project Guide	Project Co-Ordinator	Head of Department	Principal

Name:	**Signature**
1. ……	……
2. ……	……
3. ……	……

Basaveshwar Engineering College (Autonomous), Bagalkot

Department of Electrical and Electronics Engineering

Outcome Based Learning

Note: Mini Project is an important integral part of BE (EEE) program. Mini Project is outcome 3 of years of engineering program and is expected to test the learning skills of a student. It reflects quality of teaching–learning process in the department. Project work will remain as an epitome of your entire professional career.

(For internal audit of teaching learning outcome)

Sl. No.	Project Attributes	Mention the curricular / co-curricular / extracurricular activity that assisted	CO	PO
1	Literature Survey	1. Design and Analysis of DC–DC Boost Converter for Solar Photovoltaic Applications – IJERA. 2. Performance Analysis of DC–DC Boost Converter for Solar PV System – IJPEDS. 3. PWM Controlled DC–DC Boost Converter for Solar Energy Applications – IJAREEIE. 4. Analysis and Design of Inductor for High Frequency DC–DC Converters – IEEE Transactions on Power Electronics.	1	1
2	Problem Formulation and Objectives	Solar photovoltaic panels produce variable and low-level DC voltage depending on irradiation and temperature, which cannot be directly used for most DC loads requiring stable higher voltage. Therefore, a DC–DC boost converter is required to step up and regulate the solar panel output. The objective is to design a PWM-controlled boost converter to obtain a stable 24 V DC output from a low and fluctuating solar input.	2,4	3,4
3	Methodology and Implementation	1. Study of solar panel electrical characteristics. 2. Selection of boost converter topology.	3,5	5,6

		3. Calculation of duty cycle, switching frequency, inductor, and capacitor values. 4. PWM generation using IC555 timer.		
4	Results and Discussions – Inference	1. Successful boosting of solar voltage from ~17–18 V to regulated 24 V DC. 2. Stable PWM operation at ~100 kHz. 3. Verification of calculated inductor value for continuous conduction mode. 4. Reduction of output ripple using proper filtering.	8	7,12
5	Thesis Writing	Effective technical documentation and report preparation for the boost converter design, analysis, and experimental validation.	8	9,10,11
6	Planning Budget and Procurement	Selection and procurement of power electronic components such as MOSFET, diode, inductor, capacitors, and IC 555 timer, and planning of hardware cost and testing resources.	5,7	4
7	Project Outcome: Technical Publication	The project demonstrates a practical low-cost solar DC–DC boost converter suitable for small-scale renewable energy systems and educational applications.	8	9,11
8	Presentation – Viva Voce	Internal and external project presentation, technical discussion, and viva-voce examination demonstrating understanding of solar power electronics and boost converter design.	7,8	10,11

Mini Project COs

At the end of this course students should be able to,

1. Identify, formulate & analyze the engineering problems associated with electrical & electronics engineering and interdisciplinary research.

2 Design & implement proposed solutions for complex engineering problems to meet specified objectives by analyzing / validating the design / solutions of engineering problems using contemporary tools & resources. Identify optimization opportunities to enhance reliability and sustainability.

3. Prepare engineering documents and make effective presentation to communicate effectively and collaboratively with detailed analysis and interpretation of results to yield valid conclusions.
4. Demonstrate social, ethical cultural & engineering professional responsibilities.

PROGRAM OUTCOMES

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conductive estimations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** understand the impact of the professional engineering solutions in societal and environmental contexts, demonstrate the knowledge of, and need for sustainable development.
8. **Ethics**: Apply ethical principles and commit to professional ethics, responsibilities, and norms of the engineering practice.
9. **Individual and teamwork**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Acknowledgment

Gratitude takes three forms- “A feeling from heart, an expression in words and a giving in return”. We take this opportunity to express our feelings.

We express our sincere and humble gratitude to our project guide**, Dr. R. L. Naik,** for his valuable guidance and encouragement given during the course of the project work.

We are thankful to **Dr. R. L. Naik** HOD, Department of Electrical and Electronics Engineering for providing the facilities in the department to do project work.

We express our sincere and humble gratitude to **Dr. B. R. Hiremath** Principal, Basaveshwar Engineering College (Autonomous), Bagalkote for providing a congenial environment in the college.

We would like to convey our thanks to the Project Coordinator **Prof. S. S. Kandagal**, and we extend our sincere requital to all the faculty and staff of the Electrical and Electronics Department.

A heart full thanks to our parents and to all our family members whose patience and kind cooperation at all the stages enabled us to complete the dissertation successfully.

Project Associates

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ABSTRACT

Solar photovoltaic (SPV) systems have progressed significantly since the discovery of the photovoltaic effect in 1839 by Edmond Becquerel, with key advancements such as the development of the first practical silicon solar cell at Bell Labs in 1954 and large-scale adoption during the energy crises of the 1970s. Today, SPV technology is widely used as a clean and renewable energy source in applications ranging from rural electrification to grid-connected systems, supported by continuous improvements in efficiency and reductions in cost. Despite these advancements, small-scale SPV systems still face challenges due to the fluctuating output voltage of solar panels caused by variations in solar irradiation, temperature, and load conditions.

A major issue is that the typical maximum power point voltage (Vmp) of low-power solar panels (10–20 W) is around 17 V, which is insufficient for applications requiring a stable higher DC voltage such as 24 V systems. To address this, DC–DC boost converters are commonly used, with the conventional boost topology being preferred due to its simplicity, low component count, and compatibility with MPPT techniques. In this project, a boost converter is designed and developed for a 10 W SPV system to step up the panel voltage from approximately 17 V to a regulated 24 V DC. The converter uses a 555 timer–based PWM control operating at about 100 kHz with a duty cycle of 0.3, along with a calculated 280 µH inductor and suitable passive components. Experimental results demonstrate stable voltage regulation and reliable performance, providing a low-cost and effective solution for small-scale solar applications with minimal complexity.

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Dedicated

to

Our Parents

Teachers

and

Friends

Chapter-1
Introduction

1.1 Introduction

The increasing demand for renewable energy sources has led to extensive use of solar photovoltaic (SPV) systems for small- and medium-scale power generation. Solar PV panels convert sunlight directly into electrical energy; however, the output voltage obtained from a solar panel is inherently variable and depends on factors such as solar irradiation, temperature, and load conditions. This variable nature of the solar output makes it unsuitable for directly powering most electronic devices and DC loads that require a stable and regulated voltage supply.

DC–DC power converters play a vital role in conditioning the electrical energy generated by solar panels. Among the various DC–DC converter topologies, the boost converter is widely used when the required output voltage is higher than the available input voltage. A boost converter operates by storing energy in an inductor during the switching ON period and releasing it to the load during the switching OFF period, thereby increasing the output voltage above the input level.

In this project, a boost converter is designed and developed to step up the solar panel voltage of approximately 17–18 V to a regulated ~24 V DC output. The switching operation is controlled using a pulse-width modulation (PWM) signal generated by a IC 555timer integrated circuit. Proper selection of switching frequency, duty cycle, and power components such as the MOSFET, diode, capacitors, and inductor is essential to achieve efficient and stable operation.

A key aspect of this project is the design and fabrication of a custom inductor suitable for high-frequency operation. The inductor value is calculated based on input voltage, load current, switching frequency, and allowable ripple current to ensure continuous conduction mode and reduced output ripple. The performance of the designed boost converter is experimentally evaluated under varying input and load conditions to verify its effectiveness.

The developed system demonstrates a low-cost and practical approach for regulating and boosting solar panel voltage, making it suitable for small renewable energy applications such as DC loads, LED lighting systems, and portable solar-powered devices.

1.1.1 Solar Photovoltaic (SPV) System

Solar energy is one of the most widely used renewable energy sources due to its abundance and eco-friendly nature. A solar photovoltaic (SPV) system converts sunlight directly into electrical energy using photovoltaic cells. These cells are made of semiconductor materials such as silicon, which generate DC voltage when exposed to sunlight due to the photovoltaic effect.

The output voltage and current of a solar panel vary with solar irradiation, temperature, and load conditions. Because of this variation, the voltage produced by the solar panel is not constant and cannot be directly used for most electrical applications. Therefore, power conditioning circuits are required to regulate and control the output voltage of the SPV system.

1.1.2 Need for DC–DC Converters in SPV Systems

DC–DC converters are an essential part of solar photovoltaic (SPV) systems because the electrical output obtained from a solar panel is inherently variable and unstable. The voltage generated by a solar panel depends on environmental factors such as solar irradiation, temperature, and load conditions. Due to continuous changes in these parameters, the panel output voltage fluctuates throughout the day, making it unsuitable for direct connection to most DC loads or power electronic systems.

In addition to voltage instability, there is often a mismatch between the voltage level produced by the solar panel and the voltage required by the load. Typically, SPV panels produce low DC voltages in the range of 12 V to 24 V, whereas many applications such as battery charging systems, DC motor drives, and inverter inputs require a higher and regulated DC voltage. A DC–DC converter is therefore required to step up or step down the panel voltage to the desired level.

DC–DC converters also play a vital role in maintaining a regulated output voltage. This ensures reliable operation of the connected load and improves the overall performance of the SPV system. In SPV applications, boost converters are commonly used to increase the fluctuating panel voltage to a stable higher DC voltage.

Furthermore, direct connection of a solar panel to a load often results in poor power transfer and reduced efficiency. DC–DC converters help in matching the impedance

between the solar panel and the load, enabling improved power transfer and higher system efficiency.

1.1.3 Types of DC–DC Converters

DC–DC converters are broadly classified based on their voltage conversion capability:

- **Buck Converter:** Steps down the input DC voltage to a lower output voltage

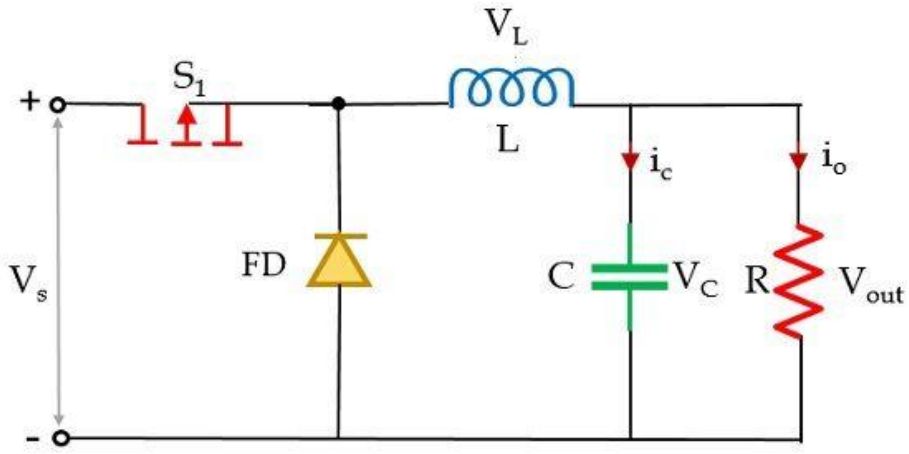


Fig 1.3.1: Buck Converter

A buck converter is a type of DC–DC converter used to step down a higher input DC voltage to a lower and regulated output voltage. It operates using a high-frequency switching device along with an inductor, diode (or synchronous switch), and capacitor. When the switch is ON, energy is stored in the inductor, and when it is OFF, the stored energy is delivered to the load through the diode, resulting in a reduced output voltage. By controlling the duty cycle of the switch, the output voltage can be precisely regulated.

- **Boost Converter:** Steps up the input DC voltage to a higher output voltage

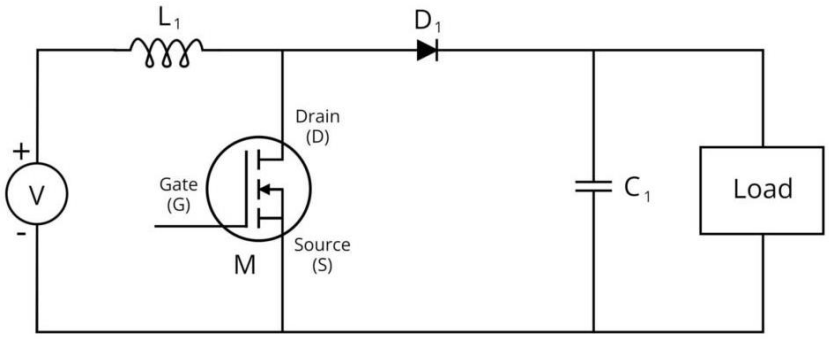


Fig 1.3.2: Boost converter

A boost converter is a type of DC–DC converter used to step up a lower input DC voltage to a higher and regulated output voltage. It operates using a high-frequency switching device, inductor, diode, and output capacitor. When the switch is ON, energy is stored in the inductor, and when the switch is OFF, the stored energy is transferred to the output through the diode, resulting in an increased output voltage. The output voltage level is controlled by adjusting the duty cycle of the switching signal.

- **Buck–Boost Converter:** Can either increase or decrease the voltage

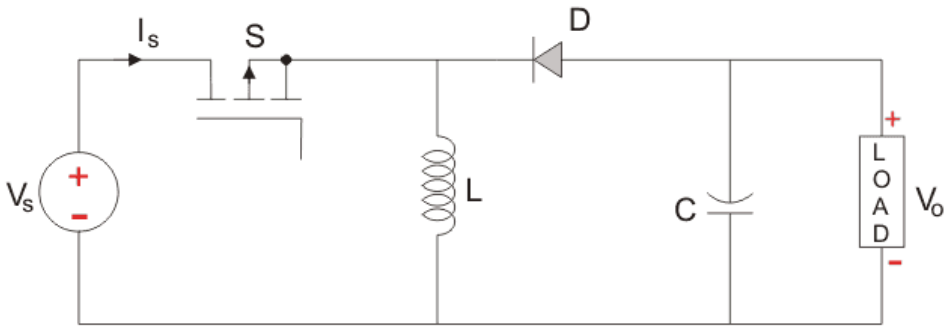


Fig 1.3.3: Buck-Boost Converter

A buck–boost converter is a type of DC–DC converter that can either step up or step down the input DC voltage depending on the requirement. It uses a switching device, inductor, diode, and capacitor to store and transfer energy. By controlling the duty cycle of the switch, the converter can produce an output voltage that is higher or lower than the input voltage**.** In the conventional buck–boost converter, the output voltage is inverted in polarity with respect to the input.

Among these, the boost converter is widely used in SPV systems where the solar panel output voltage is lower than the required load voltage.

1.2 Literature Survey

[1] Aakanksha Jain, et al., "PV with DC–DC Boost Converter and MPPT for Grid-Connected Photovoltaic Application System," ***International Journal of Engineering Research and Applications*** **(IJERA), Vol. 11, Issue 9, September 2021, pp. 12–18.**

This paper focuses on the design and analysis of a DC–DC boost converter used to regulate the variable output voltage of a solar photovoltaic (SPV) system. The authors explain that the voltage obtained from a solar panel changes with environmental conditions such as sunlight intensity and temperature, making

direct utilization difficult. To overcome this issue, a boost converter is employed to step up the low and fluctuating solar voltage to a higher and stable DC level.

The paper discusses the operating principle of the boost converter and highlights the importance of selecting appropriate design parameters such as duty cycle, switching frequency, inductor value, and output capacitor. It emphasizes that proper inductor design and PWM control are essential for achieving efficient energy transfer and reduced voltage ripple. The analysis presented in the paper demonstrates that a well-designed boost converter can significantly improve the usability of solar power for DC loads.

[2] N. Sarmah and co-authors, "Performance Analysis of DC–DC Boost Converter for Solar Photovoltaic (PV) System with MPPT Interface," *International Journal of Power Electronics and Drive Systems* (IJPEDS), Vol. 14, No. 3, pp. 1834–1843, 2023.

This paper presents a detailed performance analysis of a DC–DC boost converter used in solar photovoltaic (PV) systems. The authors analyze how variations in solar input voltage and load conditions affect the output voltage, efficiency, and overall performance of the boost converter. The study emphasizes the importance of proper design of switching frequency, duty cycle, and passive components to achieve stable and efficient operation.

This paper was referred to understand the performance behavior of boost converters under varying solar and load conditions, which helped in evaluating and validating the results of the proposed project.

[3] S. Prema, "High Efficiency DC-DC Converter for Renewable Energy Applications with High Voltage Gain and Reduced Losses," *IOSR Journal of Electrical and Electronics Engineering* (IOSR-JEEE), Vol. 12, Issue 2, Mar.–Apr. 2017, pp. 66–79.

The project Design of High Efficiency DC–DC Boost Converter for Renewable Energy System focuses on developing an efficient power conversion interface for renewable energy sources such as solar photovoltaic systems. Renewable energy sources produce variable and low-level DC voltages, which are not directly suitable for most DC loads or energy storage systems.

To address this issue, a high-efficiency DC–DC boost converter is designed to step up the input voltage to a higher, regulated DC level using pulse-width modulation (PWM) control. The design emphasizes reduced power losses, high-frequency operation, and optimal component selection to achieve efficient and stable performance.

[4] Zhe Zhang and Michael A. E. Andersen, "High Frequency AC Inductor Analysis and Design for Dual Active Bridge (DAB) Converters," *Proceedings of the IEEE Applied Power Electronics Conference* (APEC), 2016

This paper focuses on the analysis and design of inductors used in high-frequency DC–DC converters. Inductors play a crucial role in energy storage and transfer in switching converters, and their performance significantly affects efficiency, ripple current, and overall converter operation. The study discusses key inductor design parameters such as inductance value, core material selection, operating frequency, current ripple, and losses. Special attention is given to minimizing core losses and copper losses at high switching frequencies. The paper provides design guidelines and analytical methods to achieve compact size, high efficiency, and reliable operation of DC–DC converters.

Summary of Literature Survey

The reviewed literature collectively emphasizes the critical role of DC–DC boost converters in solar photovoltaic (SPV) systems for achieving voltage regulation,

improved efficiency, and reliable power delivery. The studies highlight that the variable and low-level output voltage of solar panels necessitates the use of boost converters to obtain a stable and higher DC output suitable for practical applications. Proper selection of design parameters such as duty cycle, switching frequency, inductor, and capacitor values is shown to be essential for efficient energy transfer and reduced voltage ripple. Performance analyses under varying solar irradiation and load conditions demonstrate the importance of optimized converter design for stable operation. Additionally, high-efficiency converter designs with reduced losses and high voltage gain are discussed for renewable energy applications. The literature also stresses the significance of inductor design in high-frequency DC–DC converters, as inductor characteristics greatly influence efficiency, current ripple, and overall converter performance. Together, these works provide a strong theoretical and practical foundation for the design and development of a DC–DC boost converter for SPV systems.

Chapter-2

Objectives and Methodology

2.1 Objectives

- To design and develop a DC–DC boost converter for a solar photovoltaic (SPV) system.
- To convert the variable output voltage of the solar panel into a stable and higher DC voltage suitable for DC loads.
- To boost the solar panel voltage from approximately 17–18 V to a regulated 24V DC output.
- To implement a pulse-width modulation (PWM) based control technique for voltage regulation.

2.2 Methodology

Circuit Diagram

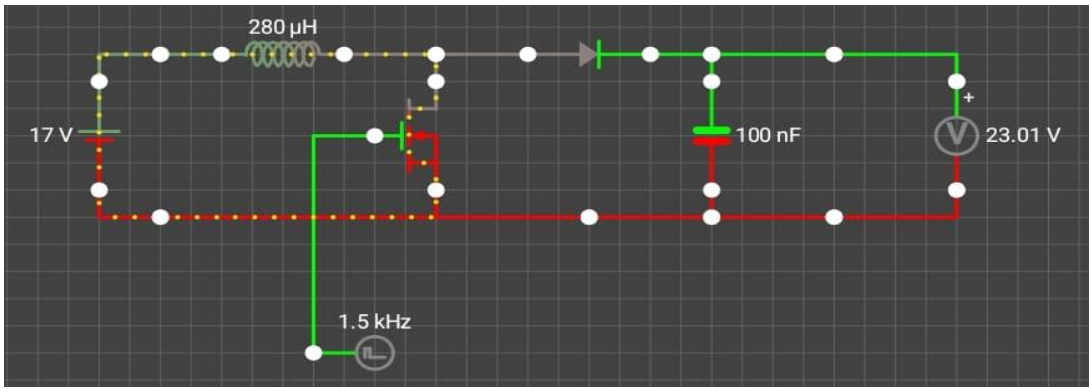


Fig. 3.2.1: Circuit diagram

The methodology adopted in this project follows a systematic approach beginning with the analysis of the electrical characteristics of the solar photovoltaic panel to determine the input voltage range and power rating. Based on these parameters, the required output voltage and power level were defined. A DC–DC boost converter topology was then selected to step up the input voltage to the desired level, and design calculations were carried out to determine key parameters such

as duty cycle, switching frequency, inductance value, and output capacitance. A switching frequency of approximately 100 kHz was chosen to achieve a balance between efficiency and component size. Finally, the complete circuit was assembled and tested under different input and load conditions, and performance parameters such as output voltage stability and ripple were analyzed to validate the effectiveness of the proposed design.

Calculation

- Solar Panel ratings :
  - Voc=21.5 V.
  - Isc=0.64 A.
  - Vm=17.0 V.
  - Im=0.6 A.
  - Pm=10W.

- Design Specifications:
  - Switching Frequency (fs) = 100 kHz
  - Output Voltage (Vout) = 24 V
  - Inductor Ripple Current = 10% of input current

- Step 1: Duty Cycle Calculation
  - $D = 1 - (Vin / Vout) = 1 - (17 / 24) = 0.292$

- Step 2: Ripple Current Calculation
  - $\Delta IL = 0.1 \times 0.6 = 0.06$ A

- Step 3: Inductor Calculation
  - $L = (Vin \times D) / (\Delta IL \times fs)$
  - $L = (17 \times 0.292) / (0.06 \times 100000)$
  - $L \approx 280$ µH

Chapter-3
Results and Discussions

3.1 Results

The boost converter was designed based on theoretical calculations to meet the required output specifications. The inductor value required for stable operation was calculated as approximately 280 µH using standard boost converter equations considering input voltage, switching frequency, duty cycle, and allowable ripple current. Using this design basis, the converter was able to boost the solar panel voltage from approximately 17–18 V to a regulated 24 V DC output.

A stable PWM signal of about 100 kHz was generated using a 555 timer IC, enabling proper switching operation of the converter. The output voltage remained stable under normal operating conditions.

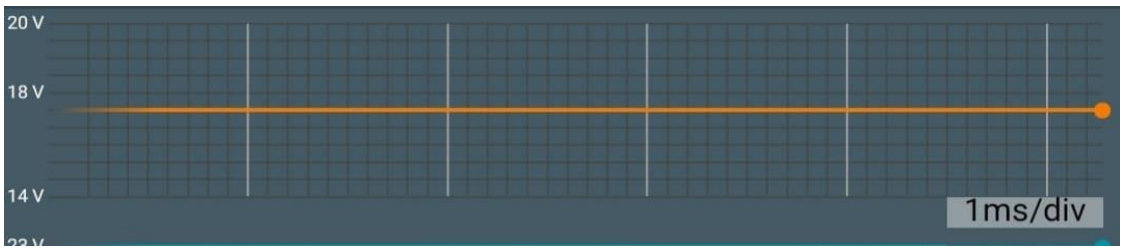


Fig. 4.1.1: Graph of Input Voltage

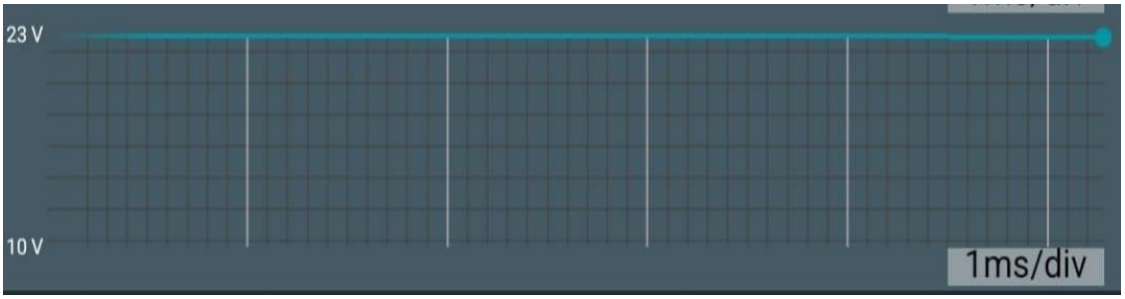


Fig. 4.1.2: Graph of Output Voltage

3.2 Discussions

This result indicates that the inductor value obtained through theoretical calculations is appropriate for the correct operation of the DC–DC boost converter. The selected switching frequency and PWM control strategy ensured that the converter operated in continuous conduction mode, which is essential for stable operation and efficient energy transfer. The boost converter was able to effectively increase the solar panel voltage to the desired level, demonstrating the validity of the design.

Small variations in the output voltage were observed, mainly due to fluctuations in the solar panel input caused by changes in solar irradiation and operating conditions. Such variations are normal and unavoidable in practical solar photovoltaic systems. Although the inductor was not physically designed or fabricated, the calculated inductance value proved sufficient for selecting a suitable commercially available inductor. This confirms that the theoretical design methodology used in the project is reliable and can be confidently applied for practical boost converter implementation in SPV systems.

Chapter-4

Conclusion

Conclusion

This project successfully demonstrated the design and implementation of a DC–DC boost converter for a solar photovoltaic system to step up the variable output voltage of a solar panel to a stable and higher DC level. The converter was designed to boost the solar panel voltage of approximately 17–18 V to a regulated 24 V DC output using PWM control generated by a 555 timer IC. Theoretical calculations were carried out to determine key design parameters such as duty cycle, switching frequency, and inductance, ensuring proper and stable operation of the converter.

The experimental results confirmed that the boost converter effectively increased the input voltage and maintained acceptable output stability under varying solar and load conditions. Although advanced features such as MPPT and closed-loop digital control were not implemented, the project clearly demonstrates the fundamental operation and practical feasibility of boost converters in solar applications. Overall, the developed system provides a simple, low-cost, and educational platform for understanding solar power conditioning and serves as a foundation for further enhancement toward real-world and commercial solar energy systems.

References

[1] Aakanksha Jain, et al., "PV with DC–DC Boost Converter and MPPT for Grid-Connected Photovoltaic Application System," *International Journal of Engineering Research and Applications* (IJERA), Vol. 11, Issue 9, September 2021, pp. 12–18.

[2] N. Sarmah and co-authors, "Performance Analysis of DC–DC Boost Converter for Solar Photovoltaic (PV) System with MPPT Interface," *International Journal of Power Electronics and Drive Systems* (IJPEDS), Vol. 14, No. 3, pp. 1834–1843, 2023.

[3] S. Prema, "High Efficiency DC-DC Converter for Renewable Energy Applications with High Voltage Gain and Reduced Losses," *IOSR Journal of Electrical and Electronics Engineering* (IOSR-JEEE), Vol. 12, Issue 2, Mar.–Apr. 2017, pp. 66–79.

[4] Zhe Zhang and Michael A. E. Andersen, "High Frequency AC Inductor Analysis and Design for Dual Active Bridge (DAB) Converters," *Proceedings of the IEEE Applied Power Electronics Conference* (APEC), 2016

[5] Kummara Venkata Guru Raghavendra, Kamran Zeb, Anand Muthusamy, T. N. V. Krishna, S. V. S. V. Prabhudeva Kumar, Do-Hyun Kim, Min-Soo Kim, Hwan-Gyu Cho & Hee-Je Kim, "A Comprehensive Review of DC–DC Converter Topologies and Modulation Strategies with Recent Advances in Solar Photovoltaic Systems," *Electronics*, Volume 9, Issue 1, Article 31, 2020.

[6] Moulahcene Fateh, Rostom Khalef, Merazga Ammar, Djamel Rahem, Imad Benacer, *"Low-Cost Arduino-Based Implementation of PWM Boost Converter for Photovoltaic (PV) Systems,"* in Proceedings of the *5th International Conference on Electrical, Computer and Energy Technologies (ICECET)***,** Paris, France, July 2025.